

 OR-AS Operations Research Applications and Solutions	Case Name: Apartment Building (1)	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Submitted by	Nicolas Van Den Noortgate		User defined distributions
Date	2014	Project Control	Automatic tracking
File Name	C2014-05 Apartment Building (1)		Tracking based on user input

1. Project description

Project authenticity

The project involves constructing a residential apartment building, starting with site investigation and setup, followed by demolition, excavation, and foundation works, then progressing through structural construction, technical installations, interior finishing, and concluding with exterior landscaping and final cleaning.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	26	Serial/Parallel (SP)	58%
Planned Duration (PD)	228	Activity Distribution (AD)	71%
Budget At Completion (BAC)	532,410.29 €	Length of Arcs (LA)	35%
Renewable Resources	-	Topological Float (TF)	18%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	15	15	2.07
CRI-rho	17	16	1.82
CRI-tau	15	21	3.43

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	60	45	-0.44
SI	75	34	-1.10
SSI	11	18	3.13
CRI-r	14	17	2.97
CRI-rho	13	17	3.21
CRI-tau	10	13	3.01

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	1.90	0.57
PV - SPI	2.93	0.78
PV - SCI	4.52	0.76
ED - 1	1.87	0.54
ED - SPI	2.84	0.69
ED - SCI	3.65	0.63
ES - 1	1.42	0.44
ES - SPI(t)	2.14	0.52
ES - SCI(t)	3.05	0.48

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.83	-21.66
CPI	1.36	-20.95
SPI	2.17	0
SPI(t)	1.83	-16.24
SCI	3.26	0
SCI(t)	2.93	-21.95
0.8 CPI + 0.2 SPI	1.53	-13.96
0.8 CPI + 0.2 SPI(t)	1.44	-19.44

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 13 tracking. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-41736.44	-61471.85	-32.73	0.84	0.77	0.76	1
std dev	15266.27	49529.90	10.5	0.06	0.17	0.07	0
final	-59000.19	0.00	-46	0.90	1	0.83	1

2.3.3.2. Time forecasting

PD	228	Real Duration	274	Late/early	20.17%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	254.85	21.22	11.77	-11.77
PV - SPI	311.69	83.46	36.71	-36.71
PV - SCI	378.62	124.84	66.06	-66.06
ED - 1	262.85	17.76	15.28	-15.28
ED - SPI	319.08	77.34	39.95	-39.95
ED - SCI	361.46	117.99	58.54	-58.54
ES - 1	261	10.86	14.47	-14.47
ES - SPI(t)	303.46	27.41	33.1	-33.1
ES - SCI(t)	340.77	61.82	49.46	-49.46

2.3.3.3. Cost forecasting

BAC	532,410.29 €	Real Cost	591,410.53 €	Over/under budget	11.08%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	574146.74	15266.27	3	-3
CPI	638639.28	44097.22	8	8
SPI	707184.18	163276.55	20	20
SPI(t)	678228.19	78145.13	15	15
SCI	808637.38	262573.88	37	37
SCI(t)	770197.38	162405.84	30	30
0.8 CPI + 0.2 SPI	648319.21	56714.18	10	10
0.8 CPI + 0.2 SPI(t)	645805.04	49949.44	9	9